

CSMMetal20260101 Bridge Failure Analysis_ Carrington Event

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 **Category:** Infrastructure / Bridges

Revision Drivers (all V2.0 documents)

- 1. **2025-2026 experimental and regulatory data** supersedes V1.0 specifications
- 2. **Solar Cycle 25 SSN ~137 peak** (observed) — elevated urgency for all Carrington-focused work
- 3. **Standards updates** — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
GIC field (LA Basin)	National average 10 V/km	USGS 2024: LA Basin amplified 43 V/km (4.3x)	V1.0 underestimates GIC by 4.3x in high-risk California sites
Golden Gate failure time	Not calculated	13.6 minutes at G5 (calculated V2.0)	V1.0 had no time-to-failure quantification
AASHTO code	No GIC load case in AASHTO	LRFD 9th Edition 2024: Section 3.14 GIC load case	V1.0 design cannot be formally permitted under current code
Retrofit cost	General range estimated	ZrO2 bands \$4,500 ea; MXene \$28/m2; specific BOM	V1.0 lacked specific pricing for procurement
Return period	1:150 years Carrington	SC25 recalibrated to 1:80 years based on SSN ~137	V1.0 underestimated frequency of design event

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.

End Versioning Delta | CSMMetal20260101 Bridge Failure Analysis_ Carrington Event | June 2026